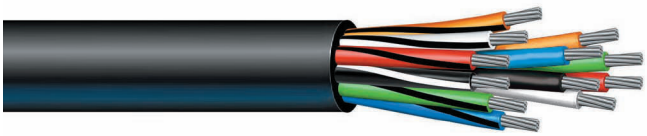
Draka

# BOSTRIG™ TYPE P CONTROL CABLE 600V OR 0.6/1kV

Multi-conductor / **unarmored**

TYPE P CONTROL CABLE 600V or 0.6/1kV **12 & 10 AWG**



## *Applications*

Bostrig™ Type P Marine and Offshore Cable is primarily designed for power, control, signal, and instrumentation applications for offshore, land drilling rigs, marine vessels, and offshore production facilities.

Bostrig™ cables have excellent resistance to oil, abrasion, moisture, vibration, sunlight, and ester based mud (Type P- MR). They are suitable for use in Class 1, Division 1 offshore applications (armored & sheathed).

The standard insulation has a continuous operating temperature of 125°C, allowing for higher ampacity levels. These cables also meet cold bend requirements of -40°C and cold impact of -35°C (CSA 22.2 NO. 0.3).

This product may be manufactured in an unarmored or armored and sheathed version.

## *Features/Ratings*

- Superior resistance to oil, abrasion, moisture, sunlight, mud, crush and impact
- High strand count conductors provide superior flexibility
- Higher allowable conductor operating temperature results in increased ampacity
- Cold bend/ cold impact of -40°/ -35°C in accordance with CSA 22.2 No. 0.3
- Flame retardant in accordance with IEEE 1202 and IEC 60332-3-22 Category A
- Arctic neoprene jacket allows ease of stripping in all temperatures
- Meets IEEE standards for 600V and performance requirements of IEC standards for 0.6/1kV
- Unarmored cables suitable for use in Class I Division 2 and Zone 2 hazardous locations
- Meets requirements for UL 1277 Type TC-ER for exposed runs

## *Approvals*

IEEE 1580 and IEEE 45- Marine Shipboard Cable

UL 1309- Marine Shipboard Cable Type X110

CSA 22.2 No. 245- Marine Shipboard Cable Type X110

CSA 22.2 No. 239- Type CIC

CSA 22.2 No. 230- Type TC-ER

Det Norske Veritas (DNV)

American Bureau of Shipping (ABS)

Transport Canada Approved AMS400-20-2

Transport Canada 8700-20-2

Lloyd's Register of Shipping (LRS)

United States Coast Guard-46CFR

## *Construction*

**CONDUCTOR:** Soft annealed stranded tinned copper per ASTM B 33. A polyester tape separator is used over the conductor.

**INSULATION:** Bostrig Type P chemically cross-linked polyolefin (XLPO), meeting IEEE 1580.

**JACKET: F**lame-retardant Arctic Neoprene, complying with Type N Neoprene as required in IEEE-1580. Thickness as shown on data sheet for unarmored version.



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### 12 AWG • 3.08 mm²

| Type Designation | Draka Number | Number of Conductor | Insulation Thickness | | Sheath Thickness | | Cable Diameter | | Cable Weight | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | in | mm | in | mm | in | mm | Lbs/Mft | Kg/Km |
| C12PN-2 | 026224 | 2 | 0.030 | 0.76 | 0.060 | 1.5 | 0.430 | 10.9 | 110 | 165 |
| C12PN-3 | 026225 | 3 | 0.030 | 0.76 | 0.060 | 1.5 | 0.450 | 11.4 | 140 | 210 |
| C12PN-4 | 026226 | 4 | 0.030 | 0.76 | 0.060 | 1.5 | 0.490 | 12.4 | 175 | 260 |
| C12PN-5 | 026227 | 5 | 0.030 | 0.76 | 0.060 | 1.5 | 0.540 | 13.7 | 210 | 315 |
| C12PN-6 | 026228 | 6 | 0.030 | 0.76 | 0.060 | 1.5 | 0.580 | 14.7 | 250 | 370 |
| C12PN-7 | 026229 | 7 | 0.030 | 0.76 | 0.060 | 1.5 | 0.580 | 14.7 | 275 | 410 |
| C12PN-8 | 026230 | 8 | 0.030 | 0.76 | 0.060 | 1.5 | 0.620 | 15.7 | 320 | 475 |
| C12PN-10 | 026231 | 10 | 0.030 | 0.76 | 0.060 | 1.5 | 0.740 | 18.8 | 390 | 580 |
| C12PN-12 | 026232 | 12 | 0.030 | 0.76 | 0.060 | 1.5 | 0.760 | 19.3 | 445 | 660 |
| C12PN-16 | 026233 | 16 | 0.030 | 0.76 | 0.080 | 2.0 | 0.890 | 22.6 | 615 | 915 |
| C12PN-20 | 026234 | 20 | 0.030 | 0.76 | 0.080 | 2.0 | 0.970 | 24.6 | 755 | 1125 |
| C12PN-24 | 026235 | 24 | 0.030 | 0.76 | 0.080 | 2.0 | 1.080 | 27.4 | 890 | 1325 |
| C12PN-30 | 026236 | 30 | 0.030 | 0.76 | 0.080 | 2.0 | 1.150 | 29.2 | 1070 | 1590 |
| C12PN-37 | 026237 | 37 | 0.030 | 0.76 | 0.080 | 2.0 | 1.240 | 31.5 | 1295 | 1925 |
| C12PN-44 | 026238 | 44 | 0.030 | 0.76 | 0.080 | 2.0 | 1.400 | 35.6 | 1540 | 2290 |
| C12PN-60 | 026239 | 60 | 0.030 | 0.76 | 0.080 | 2.0 | 1.550 | 39.4 | 2035 | 3030 |
| C12PN-91 | 026240 | 91 | 0.030 | 0.76 | 0.110 | 2.8 | 1.920 | 48.8 | 3130 | 4660 |

### 10 AWG • 5.53 mm²

| Type Designation | Draka Number | Number of Conductor | Insulation Thickness | | Sheath Thickness | | Cable Diameter | | Cable Weight | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | in | mm | in | mm | in | mm | Lbs/Mft | Kg/Km |
| C10PN-2 | 026241 | 2 | 0.030 | 0.76 | 0.060 | 1.5 | 0.490 | 12.4 | 150 | 225 |
| C10PN-3 | 026242 | 3 | 0.030 | 0.76 | 0.060 | 1.5 | 0.510 | 13.0 | 200 | 300 |
| C10PN-4 | 026243 | 4 | 0.030 | 0.76 | 0.060 | 1.5 | 0.560 | 14.2 | 250 | 370 |
| C10PN-5 | 026244 | 5 | 0.030 | 0.76 | 0.060 | 1.5 | 0.620 | 15.7 | 300 | 445 |
| C10PN-6 | 026245 | 6 | 0.030 | 0.76 | 0.060 | 1.5 | 0.680 | 17.3 | 350 | 520 |
| C10PN-7 | 026246 | 7 | 0.030 | 0.76 | 0.060 | 1.5 | 0.680 | 17.3 | 390 | 580 |
| C10PN-8 | 026247 | 8 | 0.030 | 0.76 | 0.060 | 1.5 | 0.730 | 18.5 | 445 | 660 |
| C10PN-10 | 026248 | 10 | 0.030 | 0.76 | 0.080 | 2.0 | 0.900 | 22.9 | 600 | 895 |

The current limit on these cables should be for providing control functions through relays and switching devices. The maximum current for any one conductor should not exceed the value Table 3 for three conductor cables. The average of all conductors should not exceed the limit based on the total number of conductors in the cable taken from Table 4 multiplied by the ampacity from Table 3. Three conductor or four conductor cables with three current carrying conductors may be used for continuous power.

This information is provided for reference only. Please consult the factory or your representative to confirm all engineering information.
This information is not intended to replace the information in the appropriate and applicable standard or code.
Ampacity based on 45°C ambient temperature; 95°C values based on ABS MODU Rules Table 6· 100°C values based on IEEE 45 ▪ 110°C values based on API 14F.

## TABLE 3

Three Conductor Cable, Four Conductor Cables with Three Current Carrying Conductors 45°C Ambient

| Conductor Size | | | 95°C | 100°C | 110°C | 125°C* |
|---|---|---|---|---|---|---|
| Gauge | CMA | mm² | | | | |
| 12 | 6,503 | 3.30 | 26 | 31 | 33 | 33 |
| 10 | 10,908 | 5.53 | 37 | 41 | 44 | 44 |

*125°C ampacities based on 45°C ambient in free air. Consult factory for conditions of use.

## TABLE 4

Cables with more than Four Current Carrying Conductors

| Number of Conductors | % of 3 Conductor Ampacity Values |
|---|---|
| 4-6 | 80 |
| 7-9 | 70 |
| 10-20 | 50 |
| 21-30 | 45 |
| 31-40 | 40 |
| 41-60 | 35 |
| 61 and greater | 30 |

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## 12 AWG • 3.08 mm²

| | | | | GLAND SELECTION | | | GLAND REFERENCE CHART | |
|---|---|---|---|---|---|---|---|---|
| Type Designation | Draka Number | Cable Diameter (nominal) | | Explosion Proof: Unarmored | Non-Explosion Proof: Unarmored | Non-Explosion Proof: Unarmored | Explosion Proof: (Unarmored) Hub Size Reference | Non-Explosion Proof: (Unarmored) - NPT Thread Size Reference |
| | | in | mm | | (metric) | (NPT) | | |
| C12PN-2 | 026224 | 0.430 | 10.9 | 424UB-02 | 494AB-52/ 53 | 494NE-04/ 05/ 08 | 01 = 1/2" | 03 = 1/2" - 14 NPT |
| C12PN-3 | 026225 | 0.450 | 11.4 | 424UB-02 | 494AB-52/ 53 | 494NE-04/ 05/ 08 | 02 = 1/2" | 04 = 1/2" - 14 NPT |
| C12PN-4 | 026226 | 0.490 | 12.4 | 424UB-02 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 | 03 = 3/4" | 05 = 1/2" - 14 NPT |
| C12PN-5 | 026227 | 0.540 | 13.7 | 424UB-02/ 03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 | 04 = 1" | 08 = 3/4" - 14 NPT |
| C12PN-6 | 026228 | 0.580 | 14.7 | 424UB-02/ 03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 | 05 = 1 1/4" | 10 = 3/4" - 14 NPT |
| C12PN-7 | 026229 | 0.580 | 14.7 | 424UB-02/ 03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 | 15 = 1 1/2" | 14 = 1" - 11 1/2 NPT |
| C12PN-8 | 026230 | 0.620 | 15.7 | 424UB-03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 | 06 = 2" | 15 = 1" - 11 1/2 NPT |
| C12PN-10 | 026231 | 0.740 | 18.8 | 424UB-03/ 04 | 494AB-55/ 56 | 494NE-10/ 14/ 15/ 20 | 07 = 2 1/2" | 20 = 1 1/4" - 11 1/2 NPT |
| C12PN-12 | 026232 | 0.760 | 19.3 | 424UB-04 | 494AB-55/ 56 | 494NE-10/ 14/ 15/ 20 | 08 = 3" | 21 = 1 1/4" - 11 1/2 NPT |
| C12PN-16 | 026233 | 0.890 | 22.6 | 424UB-04 | 494AB-56 | 494NE-15/ 20 | 09 = 3 1/2" | 27 = 1 1/2" - 11 1/2 NPT |
| C12PN-20 | 026234 | 0.970 | 24.6 | 424UB-04/ 05/ 15 | 494AB-56/ 57 | 494NE-15/ 20/ 21/ 27 | | 32 = 2" - 11 1/2 NPT |
| C12PN-24 | 026235 | 1.080 | 27.4 | 424UB-05/ 15 | 494AB-57 | 494NE-21/ 27 | | 38 = 2 1/2" - 8 NPT |
| C12PN-30 | 026236 | 1.150 | 29.2 | 424UB-05 /15/ 06 | 494AB-57 | 494NE-21/ 27 | | 44 = 3" - 8 NPT |
| C12PN-37 | 026237 | 1.240 | 31.5 | 424UB-05/ 15/ 06 | 494AB-57/ 59 | 494NE-21/ 27/ 32 | | 45 = 3" - 8 NPT |
| C12PN-44 | 026238 | 1.400 | 35.6 | 424UB-06 | 494AB-59 | 494NE-32 | | |
| C12PN-60 | 026239 | 1.550 | 39.4 | 424UB-06 | 494AB-59 | 494NE-32 | | |
| C12PN-91 | 026240 | 1.920 | 48.8 | 424UB-07 | 494AB-61 | 494NE-38 | | |

## 10 AWG • 5.53 mm²

| | | | | GLAND SELECTION | | |
|---|---|---|---|---|---|---|
| Type Designation | Draka Number | Cable Diameter (nominal) | | Explosion Proof: Unarmored | Non-Explosion Proof: Unarmored | Non-Explosion Proof: Unarmored |
| | | in | mm | | (metric) | (NPT) |
| C10PN-2 | 026241 | 0.490 | 12.4 | 424UB-02 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 |
| C10PN-3 | 026242 | 0.510 | 13.0 | 424UB-02 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 |
| C10PN-4 | 026243 | 0.560 | 14.2 | 424UB-02/ 03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 |
| C10PN-5 | 026244 | 0.620 | 15.7 | 424UB-03 | 494AB-53/ 55 | 494NE-05/ 08/ 10/ 14 |
| C10PN-6 | 026245 | 0.680 | 17.3 | 424UB-03/ 04 | 494AB-55 | 494NE-10/ 14 |
| C10PN-7 | 026246 | 0.680 | 17.3 | 424UB-03/ 04 | 494AB-55 | 494NE-10/ 14 |
| C10PN-8 | 026247 | 0.730 | 18.5 | 424UB-03/ 04 | 494AB-55 | 494NE-10/ 14 |
| C10PN-10 | 026248 | 0.900 | 22.9 | 424UB-04 | 494AB-56 | 494NE-15/ 20 |



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